

PE-EC703A — Embedded System

Semester VII • 3L:0T:0P • 3 Credits

Module 5 Exam Notes: Embedded System Design using PIC Microcontroller

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Introduction to Microchip PIC16 Family

PIC Microcontroller

PIC (Peripheral Interface Controller) is a family of microcontrollers manufactured by **Microchip Technology**. PIC microcontrollers are known for their RISC-based Harvard architecture, a small instruction set (<35 instructions for mid-range PIC), low cost, wide voltage range (2.0–5.5 V), and abundant peripheral integration. They are widely used in hobbyist and industrial embedded applications.

1.1 PIC Architecture Classifications

Family	Program Word	Data Bus	Examples
PIC10 (Base)	12-bit	8-bit	PIC10F200, PIC10F220
PIC12 (Base)	12-bit	8-bit	PIC12F508, PIC12F675
PIC16 (Mid-range)	14-bit	8-bit	PIC16F84A, PIC16F877A, PIC16F873
PIC18 (Enhanced)	16-bit	8-bit	PIC18F4550, PIC18F452
PIC24 / dsPIC	24-bit	16-bit	dsPIC33, PIC24FJ
PIC32	32-bit MIPS	32-bit	PIC32MX, PIC32MZ

1.2 Key Architectural Features of PIC Mid-Range (PIC16)

- **Harvard Architecture:** Separate program (Flash) and data (RAM + SFR) buses.
- **RISC Core:** Only 35 instructions; most execute in 1 instruction cycle (4 clock cycles).
- **Instruction Cycle:** $T_{cy} = 4/f_{OSC}$ (e.g., at 20 MHz: $T_{cy} = 200$ ns).
- **8-bit Working Register (W):** Equivalent to the Accumulator in 8051.
- **File Register Architecture:** All RAM, SFRs, and I/O registers accessed via “file register” addresses.
- **Pipelining:** 2-stage pipeline (Fetch + Execute) — while one instruction executes, the next is fetched.

PIC16 Key Formulas

$$T_{cy} = \frac{4}{f_{OSC}} \quad (1 \text{ instruction cycle} = 4 \text{ clock periods})$$

$$\text{Timer0 Overflow Time} = \frac{(256 - \text{TMR0}) \times \text{Prescaler} \times 4}{f_{OSC}}$$

$$\text{ADC Resolution} = \frac{V_{REF}}{2^N - 1} \quad (N = 10 \text{ bits for PIC16F873})$$

$$\text{Baud Rate (UART)} = \frac{f_{osc}}{64 \times (\text{SPBRG} + 1)} \quad (\text{Low speed mode})$$

2. PIC16F873 — Architecture and Features

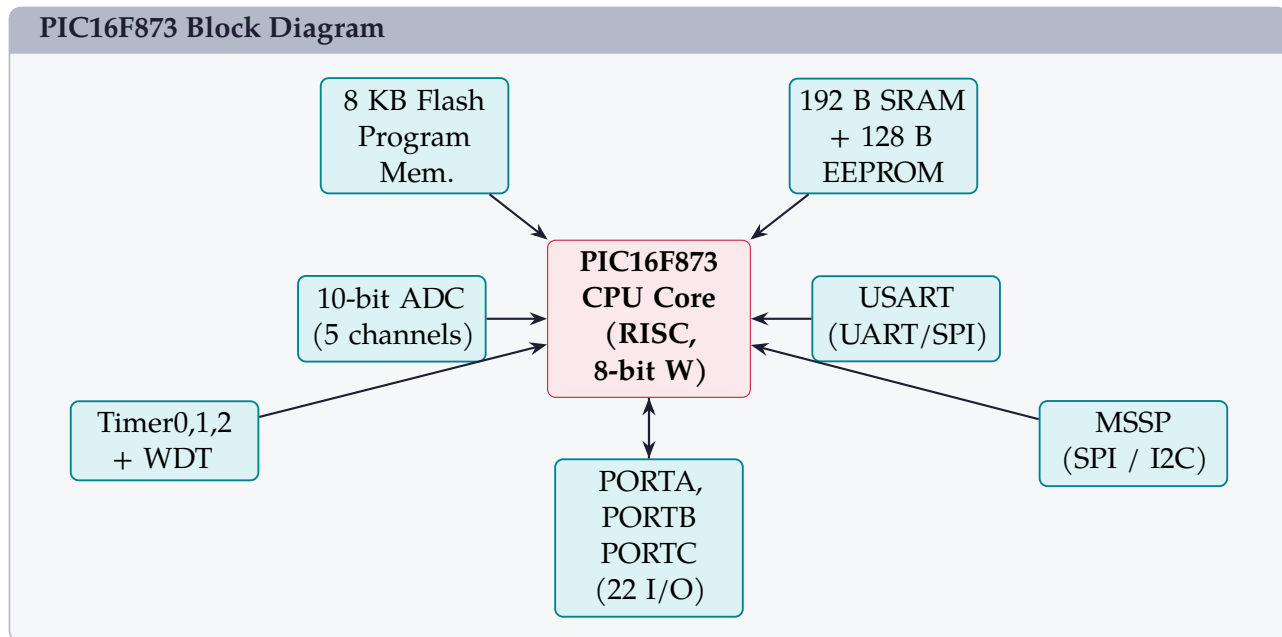
PIC16F873

The **PIC16F873** is a 28-pin Flash-based mid-range PIC microcontroller from Microchip. It is a popular choice for educational and industrial embedded designs due to its integrated analog and communication peripherals, Flash programmability, and low power consumption.

2.1 PIC16F873 — Key Specifications

Feature	Specification
Program Memory	8 KB Flash (4096 x 14-bit words)
Data Memory (RAM)	192 bytes SRAM
Data EEPROM	128 bytes
I/O Ports	PORTA (6 pins), PORTB (8 pins), PORTC (8 pins) = 22 I/O
Timers	Timer0 (8-bit), Timer1 (16-bit), Timer2 (8-bit + PR2)
ADC	5-channel, 10-bit ADC (AN0–AN4 on PORTA)
Serial Ports	USART (UART/SPI async), MSSP (SPI + I2C)
Interrupts	Up to 13 interrupt sources
Clock Speed	DC to 20 MHz (using external crystal)
Supply Voltage	2.0 V to 5.5 V
Package	28-pin PDIP, SOIC, SSOP
Watchdog Timer	Built-in WDT with independent RC oscillator

PIC16F873 Block Diagram



3. On-Chip Peripherals of PIC16F873

3.1 Watchdog Timer (WDT)

- Has its own internal **RC oscillator** (independent of main clock); continues running even if main clock fails.
- Default timeout period ≈ 18 ms; extended via **postscaler** (up to $\times 128 \approx 2.3$ s).
- Reset by `CLRWDT` instruction in software.
- On timeout: generates a device reset (MCLR pin goes low momentarily).
- Can be enabled/disabled via configuration word bits.

3.2 ADC (Analog-to-Digital Converter)

- **10-bit successive approximation ADC**: Converts analog voltage to a 10-bit digital value (0 to 1023).
- **5 analog input channels** (AN0–AN4) on PORTA (multiplexed with digital I/O).
- **Reference voltage**: V_{REF+} and V_{REF-} pins (or V_{DD} and V_{SS} by default).
- **Acquisition time**: Capacitor sampling time before conversion starts; minimum $2 \mu\text{s}$.
- **Result format**: Left-justified or right-justified; stored in ADRESH:ADRESL registers.

PIC16F873 ADC — Conversion Steps

1. Configure ADCON1 (select analog/digital pin assignment and V_{REF}).
2. Configure ADCON0 (select channel AN_x , clock source, turn ADC ON).
3. Wait for acquisition time ($\geq 2 \mu\text{s}$).
4. Set `GO/DONE` bit in ADCON0 to start conversion.

5. Poll `GO/DONE` bit until it clears (conversion complete, $\approx 12 \times T_{AD}$).
6. Read 10-bit result from `ADRESH` (high 2 bits) and `ADRESL` (low 8 bits).

3.3 Asynchronous Serial Port — USART

- Full-duplex UART on RC6 (TX) and RC7 (RX).
- Baud rate controlled by SPBRG register and BRGH bit (high/low speed).
- **High-speed mode (BRGH=1):** $\text{Baud} = f_{OSC} / (16 \times (\text{SPBRG} + 1))$
- **Low-speed mode (BRGH=0):** $\text{Baud} = f_{OSC} / (64 \times (\text{SPBRG} + 1))$
- Interrupt flags: TXIF (transmit buffer empty), RCIF (receive buffer full).

3.4 SPI and I2C — MSSP Module

- The **MSSP** (Master Synchronous Serial Port) supports both SPI and I2C modes.
- **SPI Mode:** Uses RC3 (SCK), RC4 (SDI), RC5 (SDO), and any GPIO as CS (Chip Select). Master or slave configurable.
- **I2C Mode:** Uses RC3 (SCL) and RC4 (SDA) with open-drain outputs. Supports 7-bit and 10-bit addressing.

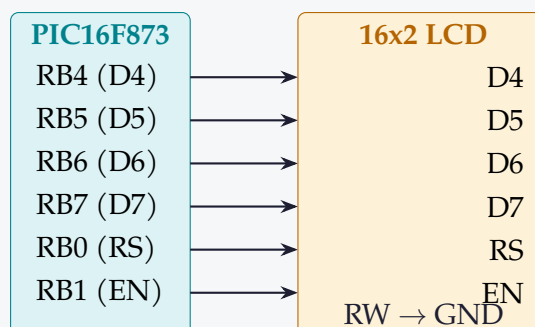
4. Interfacing with PIC16F873

4.1 LCD Interfacing

LCD (16×2) Interfacing with PIC16F873

A standard Hitachi HD44780-compatible 16×2 character LCD is interfaced using **4-bit mode** (4 data lines + 3 control lines) to save I/O pins. Control pins: RS (Register Select: 0=Command, 1=Data), RW (Read/Write, tied to GND for write-only), EN (Enable pulse to latch data).

PIC16F873 to 16×2 LCD — 4-bit Interface



4.2 ADC Interfacing (Sensors)

- Temperature sensor (LM35): Output voltage = 10 mV/°C. Connect to AN0.

– ADC result $\rightarrow$ Temperature: $T(^{\circ}C) = \frac{ADC \times V_{REF}}{1023 \times 0.01}$

- Potentiometer / Joystick: Directly connected to analog input (voltage divider).
- Light sensor (LDR): Used in voltage divider; ADC reads light level.

4.3 Stepper Motor Interfacing

- A stepper motor advances by a fixed angle (step) for each control pulse.
- **Step angle:** Depends on motor type (e.g., $1.8^{\circ}/\text{step} = 200 \text{ steps/revolution}$).
- **Driver IC:** L298N or ULN2003 connects between PIC output pins and motor coils.
- PIC sends sequence of bit patterns to PORTB to rotate the stepper.
- **Full-step sequence (4-phase):** 1100 $\rightarrow$ 0110 $\rightarrow$ 0011 $\rightarrow$ 1001 (repeat).

4.4 Keyboard (Keypad) Interfacing

- A **4 $\times$ 4 matrix keypad** uses 4 row + 4 column lines = 8 pins.
- PIC drives rows LOW one at a time, then reads which column line goes LOW.
- The row/column combination identifies the pressed key.
- Debouncing: Software delay ($\sim 20 \text{ ms}$) after detecting key press before confirming it.

4.5 DAC (Digital-to-Analog Converter)

- PIC16F873 has no built-in DAC. External options:
 - **R-2R Resistor Ladder:** Passive DAC using resistors connected to PORTB (8-bit). Simple, low cost, no extra IC.
 - **External DAC IC (MCP4921):** 12-bit DAC with SPI interface.
 - **PWM + RC filter:** Use Timer2 PWM output, filter with RC low-pass $\rightarrow$ analog voltage proportional to duty cycle.

5. Case Studies of Embedded System Design

5.1 Case Study 1: Automated Chocolate Vending Machine

Automated Chocolate Vending Machine using PIC16F873

Problem: Design a vending machine that accepts coins (detected by IR sensor), displays balance on LCD, dispenses chocolate (stepper motor) when enough coins are inserted, and provides change.

Hardware Components:

- PIC16F873 (controller), 16 $\times$ 2 LCD (display), IR coin sensor on AN0 (ADC), Stepper motor + ULN2003 (dispenser), Buzzer on RB0 (feedback), Keypad for item selection.

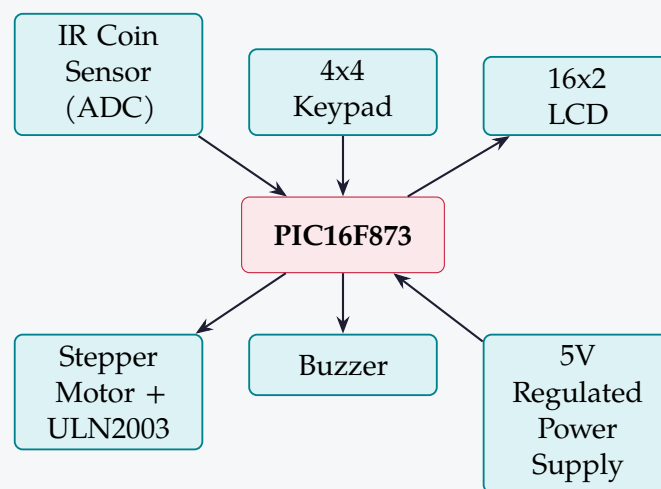
Software Flow:

1. Initialize LCD, ADC, I/O ports.

2. Display “Insert Coin” on LCD.
3. Loop: Poll coin sensor (ADC reading > threshold → coin detected).
4. On coin: Increment balance; update LCD display.
5. If balance $\geq$ price: Drive stepper motor (4 steps) to dispense; beep buzzer.
6. Calculate and dispense change (reverse stepper? No — separate change mechanism).
7. Reset balance; display “Thank You”.

Key Programming Concepts Used: ADC polling, stepper motor control sequences, LCD 4-bit mode, software debounce, timer-based delay.

Vending Machine System Block Diagram



5.2 Case Study 2: Digital Camera Controller

Digital Camera Controller using PIC (Concept)

Note: A full digital camera uses high-performance processors (e.g., ARM Cortex-A) for image processing. PIC16F873 can implement the **camera control subsystem** (not image processing).

PIC16F873 Role in Digital Camera:

- **Shutter Control:** Timer-based precise shutter speed control (1/1000s to 1/30s) using Timer1 interrupt.
- **Focus Motor Control:** Stepper motor drives lens position; PIC reads ADC from autofocus sensor.
- **Flash Trigger:** GPIO pin triggers external flash unit via optocoupler.
- **User Interface:** Reads mode dial (keypad), displays settings on small LCD.
- **Battery Monitor:** ADC reads battery voltage divider; displays charge level.
- **Serial Communication:** UART or SPI to main image processor (ARM) for command passing.

Key Techniques: Timer interrupts for precision timing, ADC for sensor reading, SPI for inter-processor communication, WDT for reliability.

Quick Revision — Module 5

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Key Formulas at a Glance

Instruction Cycle	$T_{cy} = \frac{4}{f_{OSC}}$ (at 20 MHz: $T_{cy} = 200$ ns)
Timer0 Overflow	$t_{TMR0} = \frac{(256 - TMR0_init) \times \text{Prescaler} \times 4}{f_{OSC}}$
ADC Result (10-bit)	$V_{in} = \frac{ADC \times V_{REF}}{1023}$ (right-justified)
UART Baud (BRGH=1)	$\text{Baud} = \frac{f_{OSC}}{16 \times (\text{SPBRG} + 1)}$
ADC Temperature (LM35)	$T(^{\circ}\text{C}) = \frac{ADC \times V_{REF}}{1023 \times 0.01}$

One-Line Definitions

- **PIC:** Peripheral Interface Controller — Microchip Technology's RISC-based MCU family.
- **PIC16F873:** 28-pin, 8-bit Flash PIC MCU; 8 KB Flash, 192 B SRAM, 10-bit ADC, USART, MSSP.
- **W Register:** Working Register in PIC — equivalent to Accumulator; all ALU operations use W.
- **File Register:** PIC's general-purpose RAM + SFR memory, accessed via file register addresses.
- **ADCON0/1:** ADC Control registers in PIC16F873 that configure channel selection, clock, and reference.
- **MSSP:** Master Synchronous Serial Port — supports SPI and I2C in PIC16F873.
- **USART:** Universal Synchronous/Asynchronous Receiver/Transmitter — UART + SPI on PIC.
- **SPBRG:** Serial Port Baud Rate Generator register — sets UART baud rate in PIC.
- **Prescaler:** Divides the clock signal before reaching the timer — extends the timer overflow period.
- **TMR0:** 8-bit Timer/Counter in PIC16F873 (overflows at 255 → 0, generates TMR0IF).
- **CLRWDT:** Assembly instruction to reset the Watchdog Timer in PIC.
- **Full-step sequence:** 4-phase stepper motor control pattern (1100, 0110, 0011, 1001).

Frequently Asked / Viva Questions

Q1. What does PIC stand for and who manufactures it?

PIC stands for **Peripheral Interface Controller**. It is manufactured by **Microchip Technology Inc.**

Q2. State the key features of PIC16F873.

28-pin PDIP; 8 KB Flash program memory; 192 B SRAM; 128 B EEPROM; 10-bit 5-channel ADC; three timers (Timer0/1/2); USART; MSSP (SPI + I2C); up to 20 MHz clock; 2.0–5.5 V supply; built-in WDT.

Q3. What is the W register in PIC? How is it different from 8051's Accumulator?

The **W (Working) register** is PIC's primary data register used for ALU operations. Similar to the Accumulator in 8051, it is used as the source/destination for most arithmetic and logic instructions. Unlike 8051's A, W is not directly addressable in the file register map.

Q4. What is the instruction cycle of PIC16 and how does it relate to clock frequency?

One instruction cycle = 4 clock cycles: $T_{cy} = 4/f_{OSC}$. At 20 MHz, $T_{cy} = 200$ ns. Most PIC instructions execute in 1 cycle (200 ns); branch instructions take 2 cycles.

Q5. How does the PIC16F873 ADC work? What is the resolution?

It is a 10-bit successive approximation ADC. It samples the analog voltage on AN0–AN4, converts it to a 10-bit number (0–1023) representing V_{in} between V_{REF-} and V_{REF+} . Resolution = $V_{REF}/1023$ (e.g., ≈ 4.88 mV for 5V reference).

Q6. Explain how a stepper motor is interfaced with PIC16F873.

The PIC outputs a 4-bit step sequence to PORTB (e.g., 1100, 0110, 0011, 1001 for full-step mode). Each pattern energizes the motor coils in sequence. A ULN2003 Darlington driver IC between PORTB and the motor coils amplifies the current. The step rate (delay between patterns) controls motor speed.

Q7. What is the difference between USART and MSSP in PIC16F873?

USART: Handles asynchronous serial (UART) and synchronous serial (SPI) communication on RC6/RC7. **MSSP:** A second serial module supporting both SPI (master/slave) and I2C master/slave modes on RC3/RC4/RC5. Both can be used simultaneously.

Q8. How is LCD interfaced with PIC16F873 in 4-bit mode?

Only 4 data lines (D4–D7) are connected to PORTB, along with RS and EN control lines (RW tied to GND). Data is sent as two 4-bit nibbles (high nibble first, then low nibble), each latched by pulsing EN high then low.

Q9. How does the Watchdog Timer work in PIC16F873?

The WDT has its own RC oscillator (≈ 18 ms period). Software must execute `CLRWDT` instruction before timeout. If not cleared in time (e.g., due to program hang), WDT causes a device reset. Timeout can be extended using a postscaler (up to $\times 128 \approx 2.3$ s).

Frequently Asked / Viva Questions (contd.)

Q10. What are the Timer modes available in PIC16F873?

Timer0: 8-bit timer/counter with prescaler (1:2 to 1:256). **Timer1:** 16-bit timer/counter, can use external crystal for low-power RTC. **Timer2:** 8-bit timer with prescaler and postscaler; period register PR2 sets overflow point; used for PWM generation.

Q11. Describe the design of an automated vending machine using PIC16F873.

- IR sensor (ADC) detects coins; balance tracked in SRAM variable.
- LCD (4-bit mode) shows balance and prompts.
- Keypad selects product; price compared with balance.
- If balance $\geq$ price: stepper motor (via ULN2003) dispenses product; buzzer beeps.
- Balance decremented by price; "Thank You" displayed.

Q12. Why is PIC preferred for cost-sensitive embedded applications?

PIC MCUs are very inexpensive (PIC16F84A ~ Rs. 80–160), available in small packages, have wide voltage range (2–5.5 V), abundant development tools (MPLAB IDE, free compiler), large community, and integrate all needed peripherals on-chip (eliminating external ICs). They are ideal for high-volume, cost-sensitive consumer products.

PIC16F873 vs. 8051 vs. ARM Cortex-M0 — Comparison

Feature	PIC16F873	8051 (AT89S52)	ARM Cortex-M0
Architecture	Harvard RISC (8-bit)	Harvard CISC (8-bit)	Harvard RISC (32-bit)
Instructions	35 (14-bit words)	255 (variable)	Thumb-2 (16/32-bit)
Max Clock	20 MHz	33 MHz	48 MHz
Flash	8 KB	8 KB (ISP)	32–128 KB
RAM	192 bytes	256 bytes	8–32 KB
Built-in ADC	10-bit, 5-ch	None	12-bit, 8+ ch
Price (approx.)	Rs. 40–160	Rs. 65–160	Rs. 80–400
IDE	MPLAB X	Keil, SDCC	Keil, GCC ARM